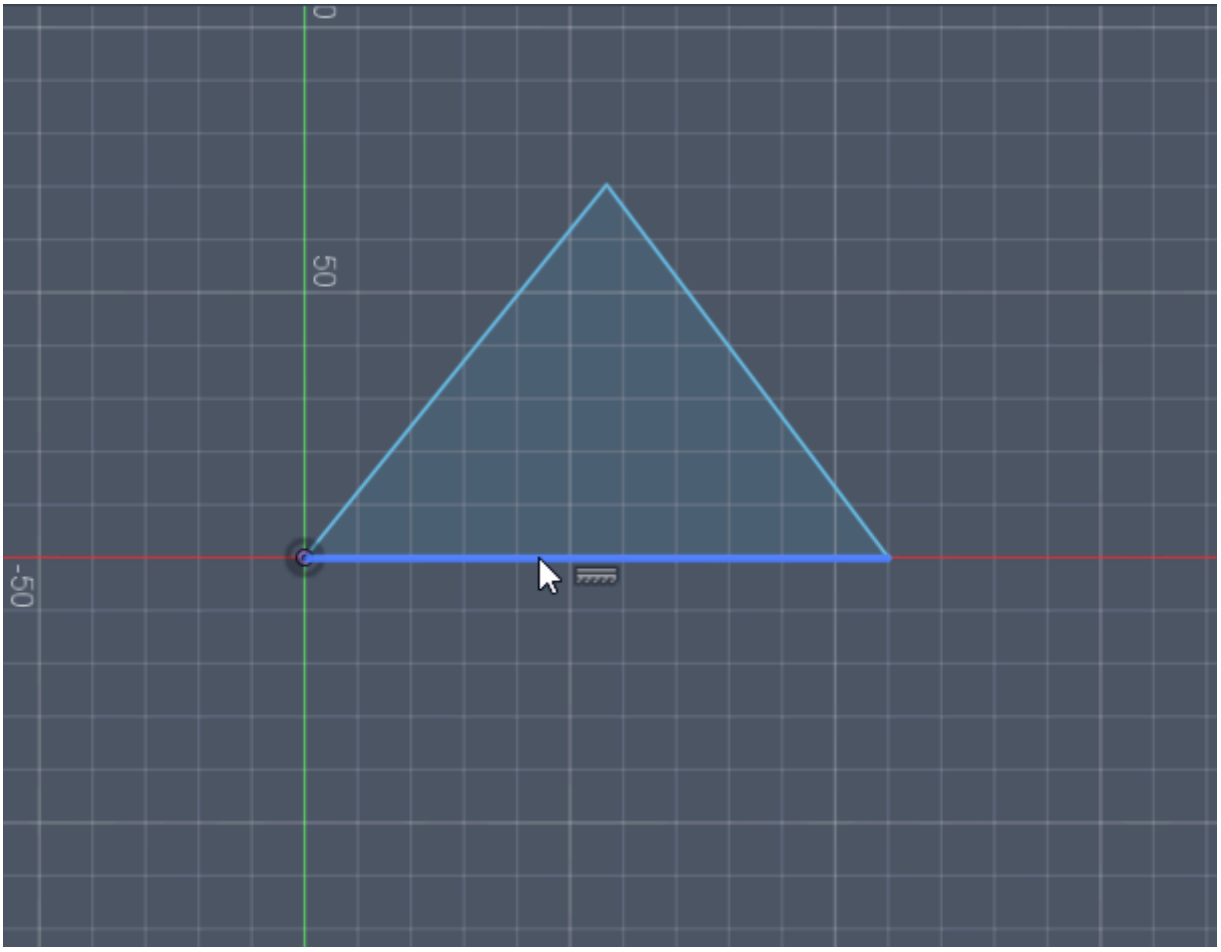
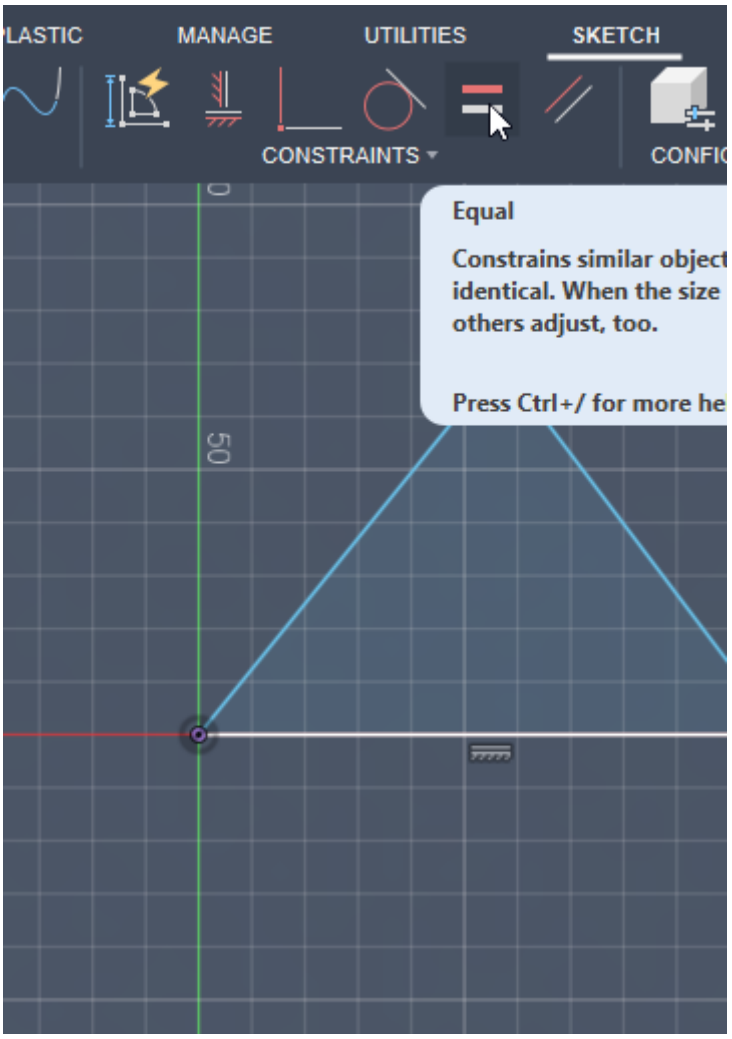
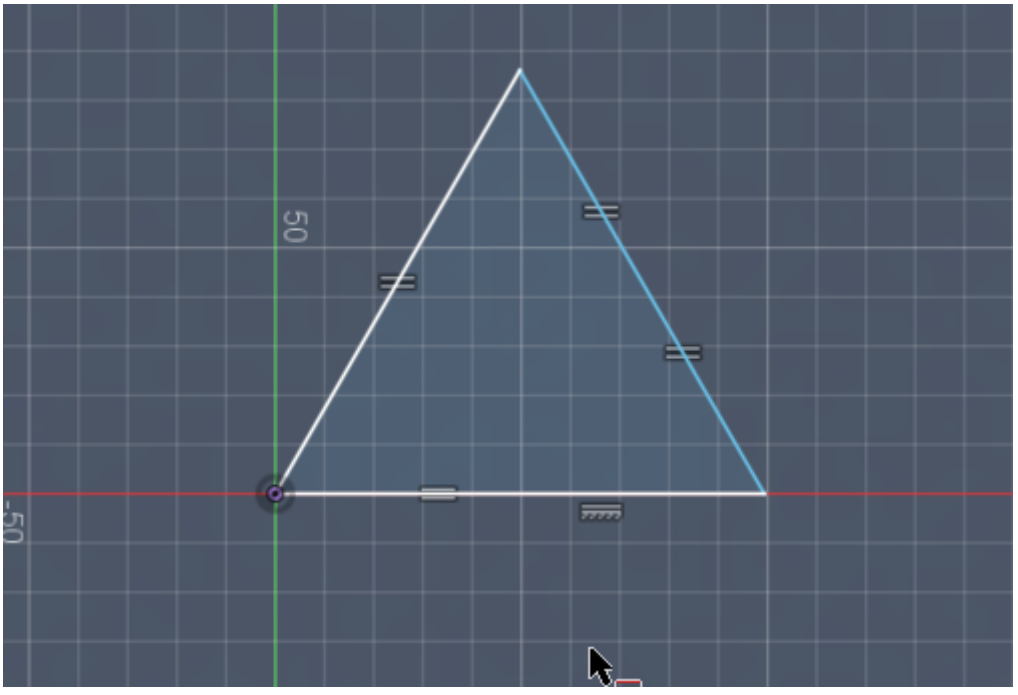
Painter’s Tripod



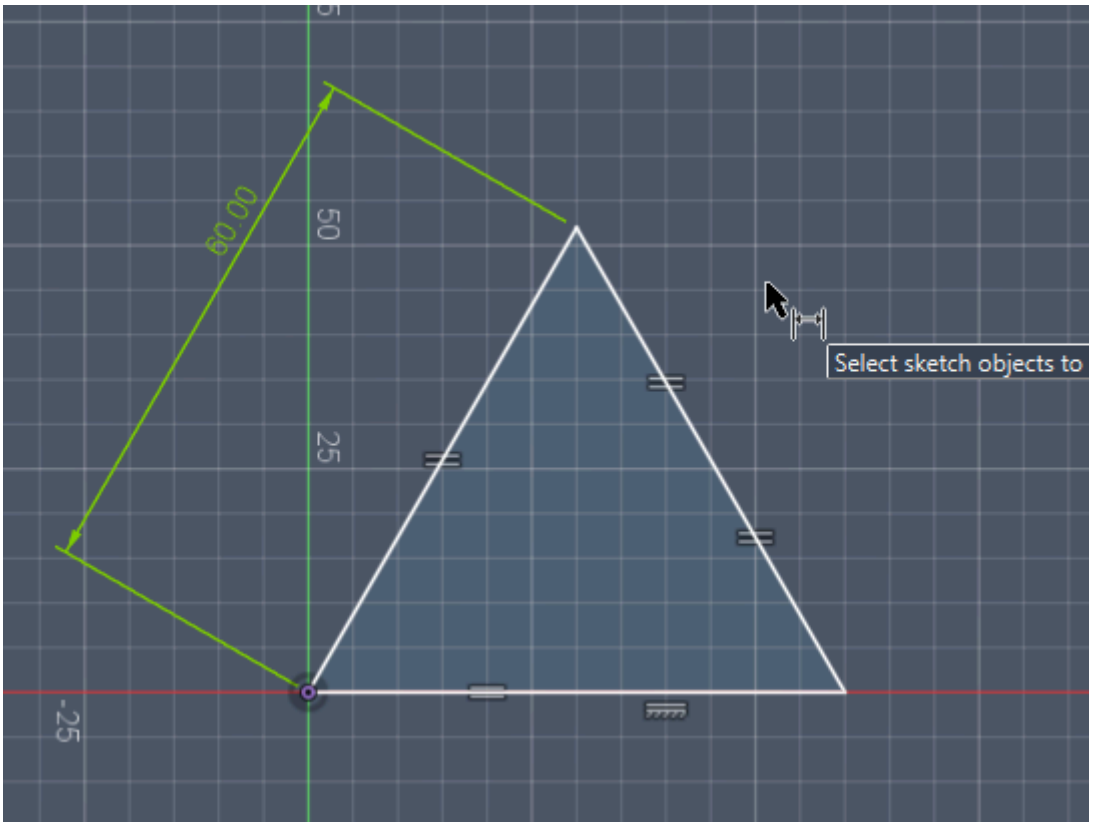
Draw a triangle first (base sketch).



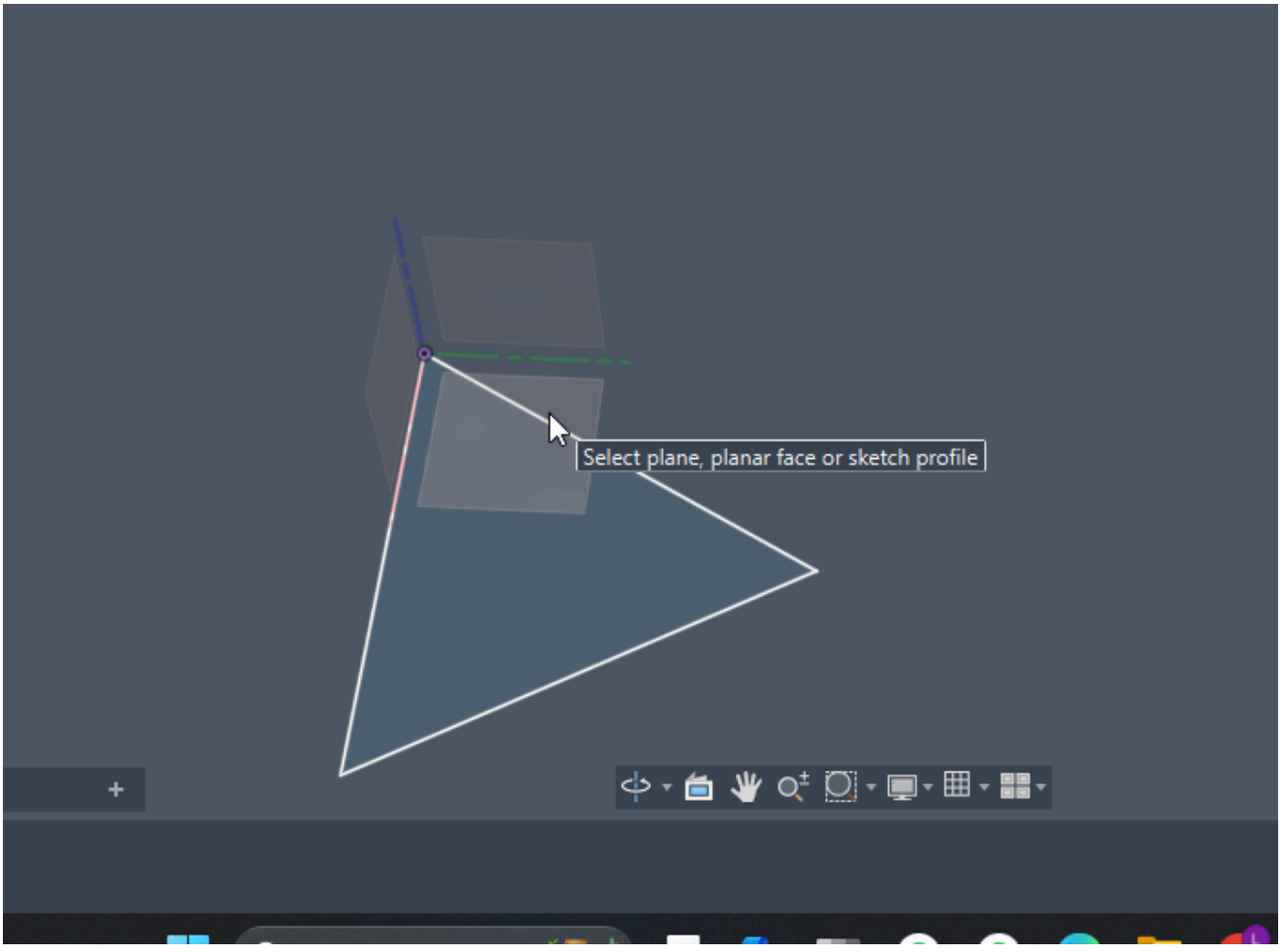
Use the Equal constraint tool to make all three sides equal.



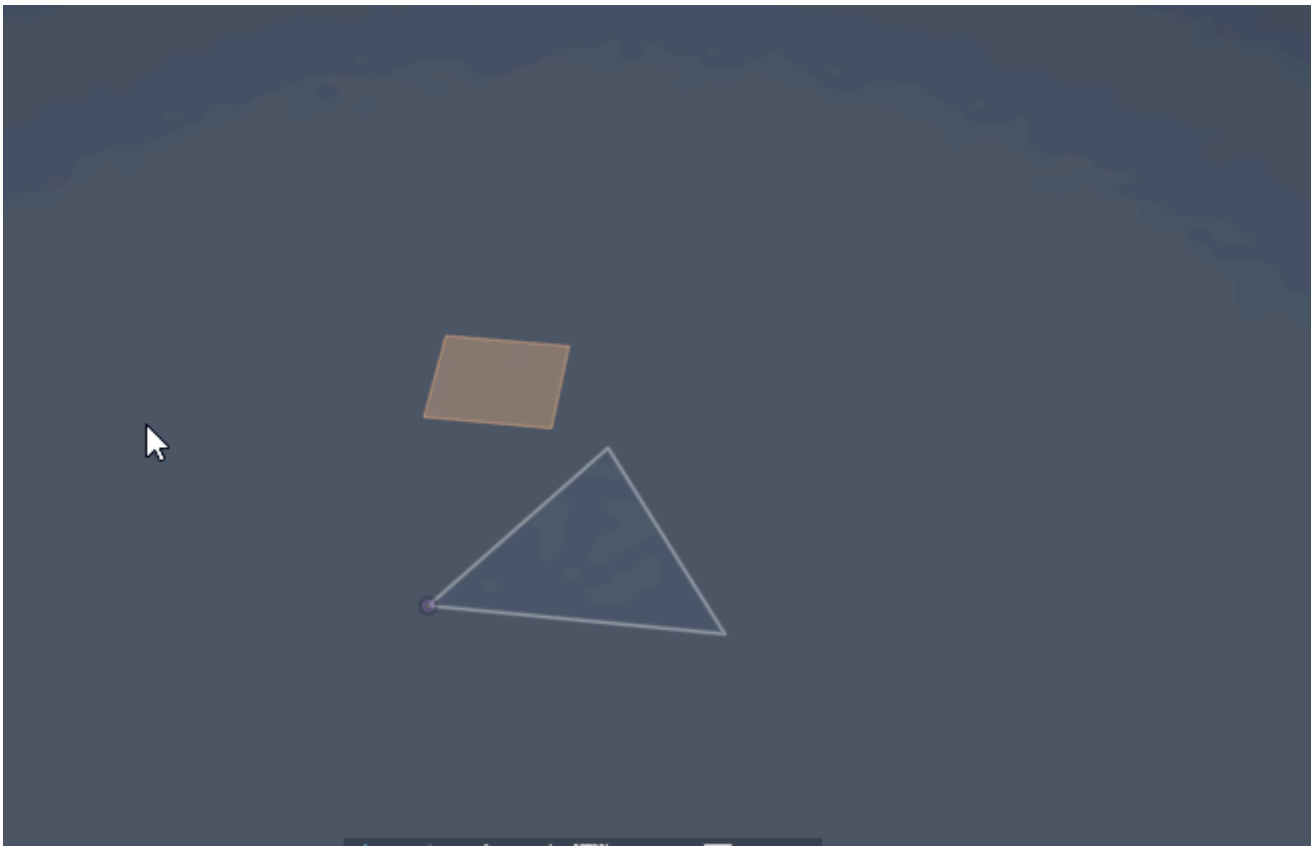
All sides become equal after applying the Equal constraint.



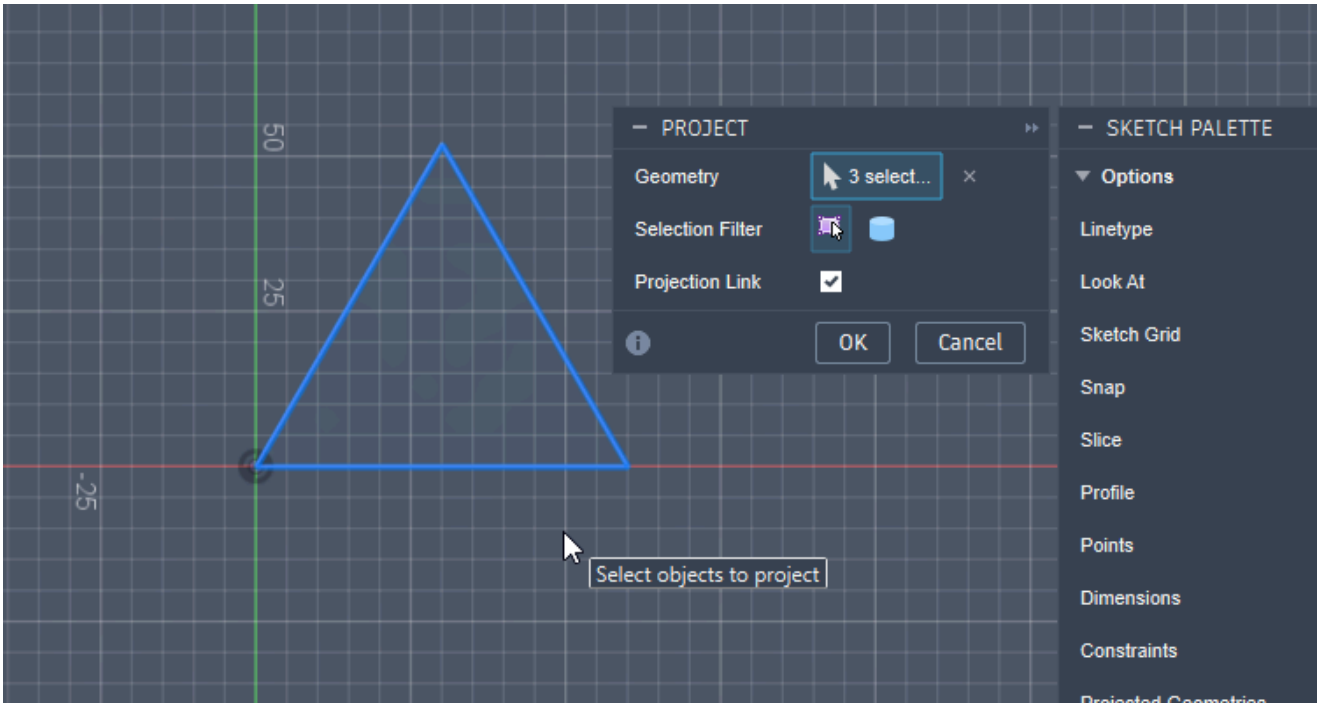
Use Sketch Dimension on one side — due to Equal, all sides update together.



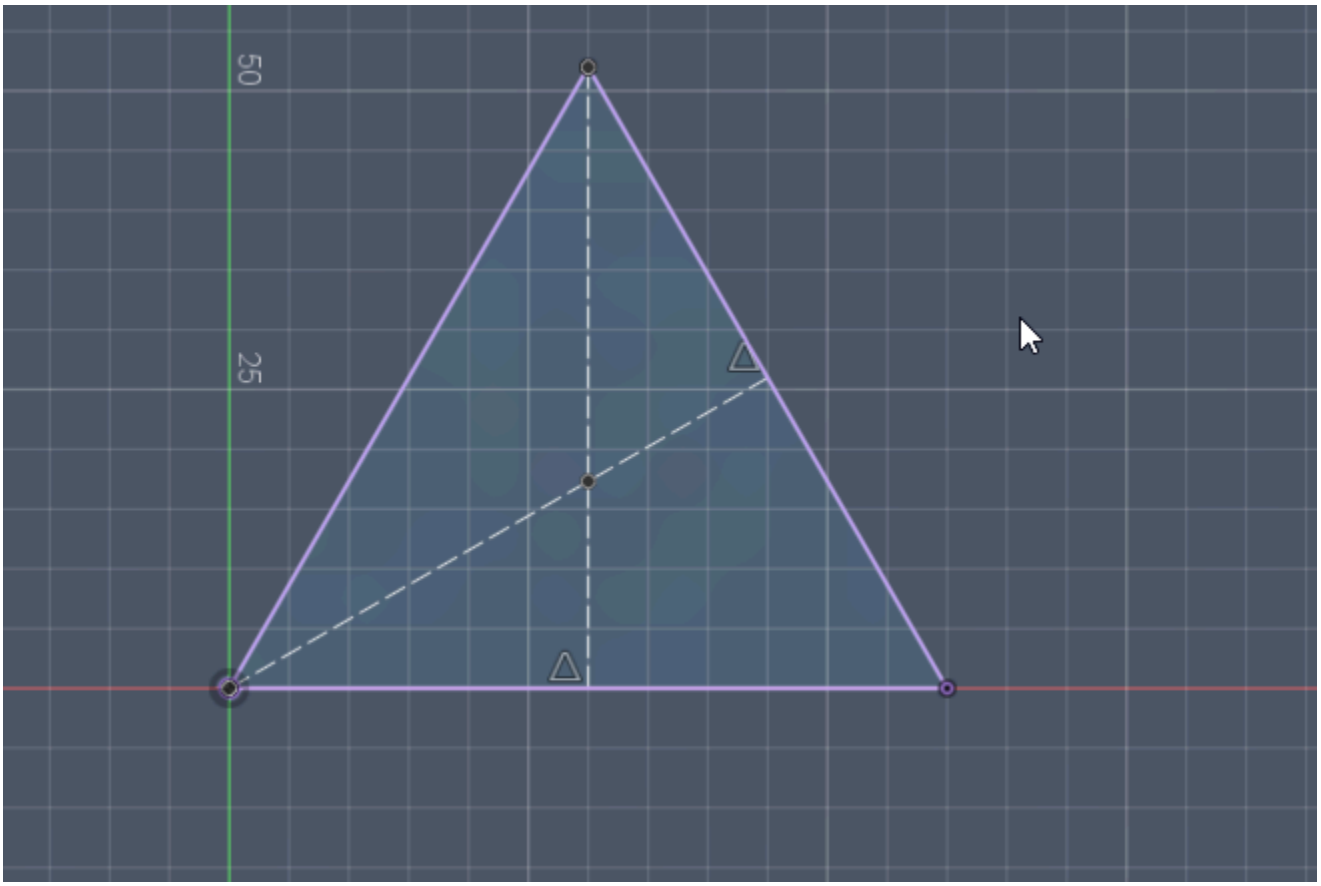
Create an Offset Plane 50 mm above the base plane.



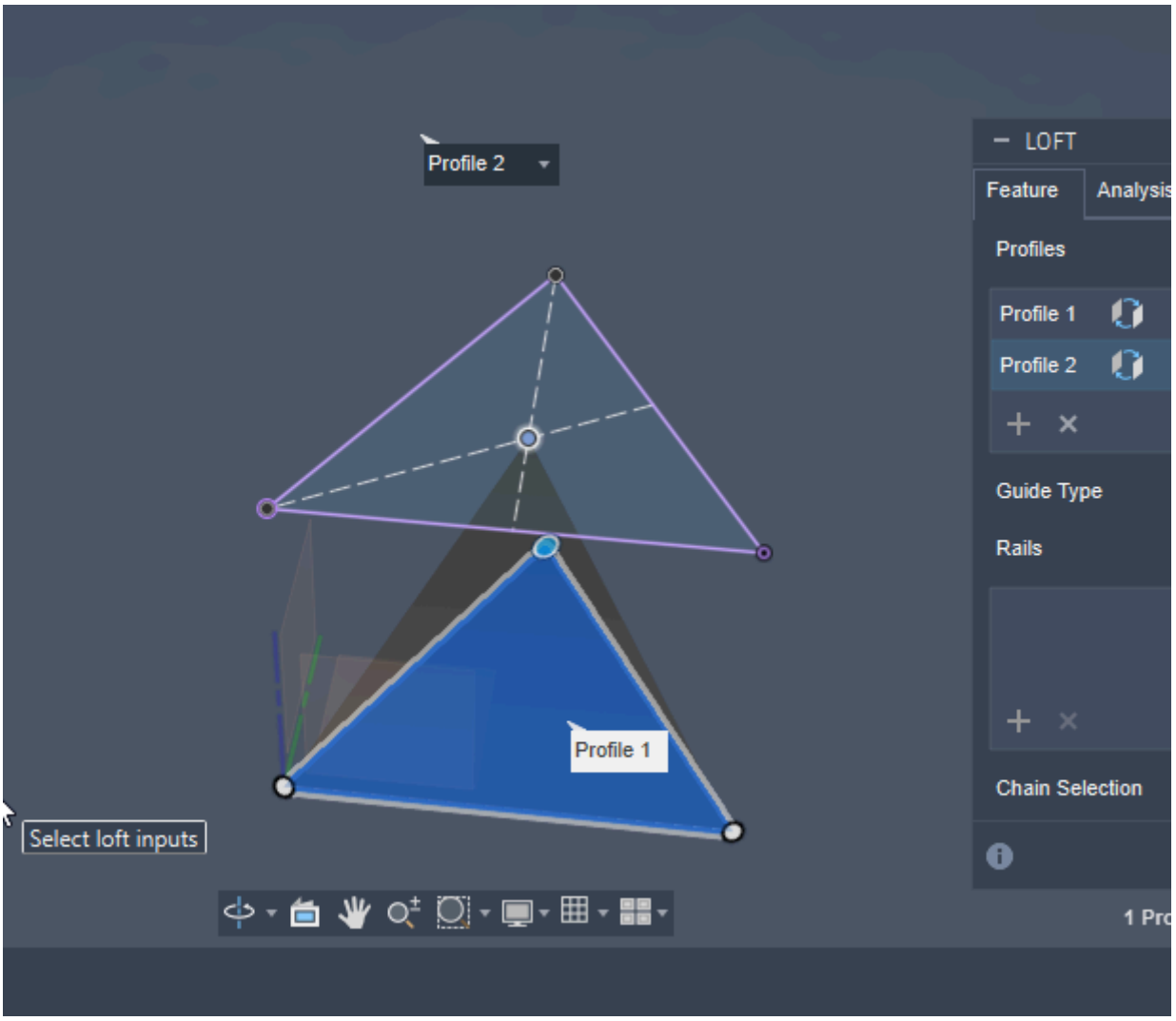
Create a new sketch on the offset plane.



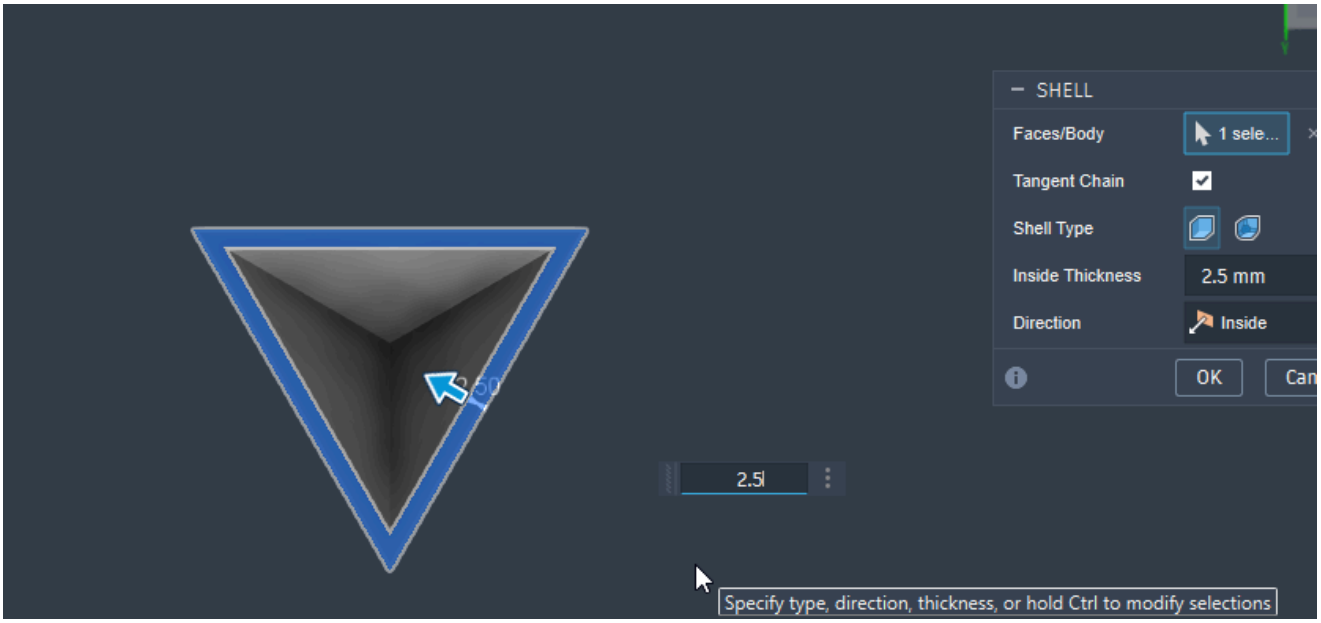
Project the triangle vertices into this sketch (keeps parametric link to main sketch).



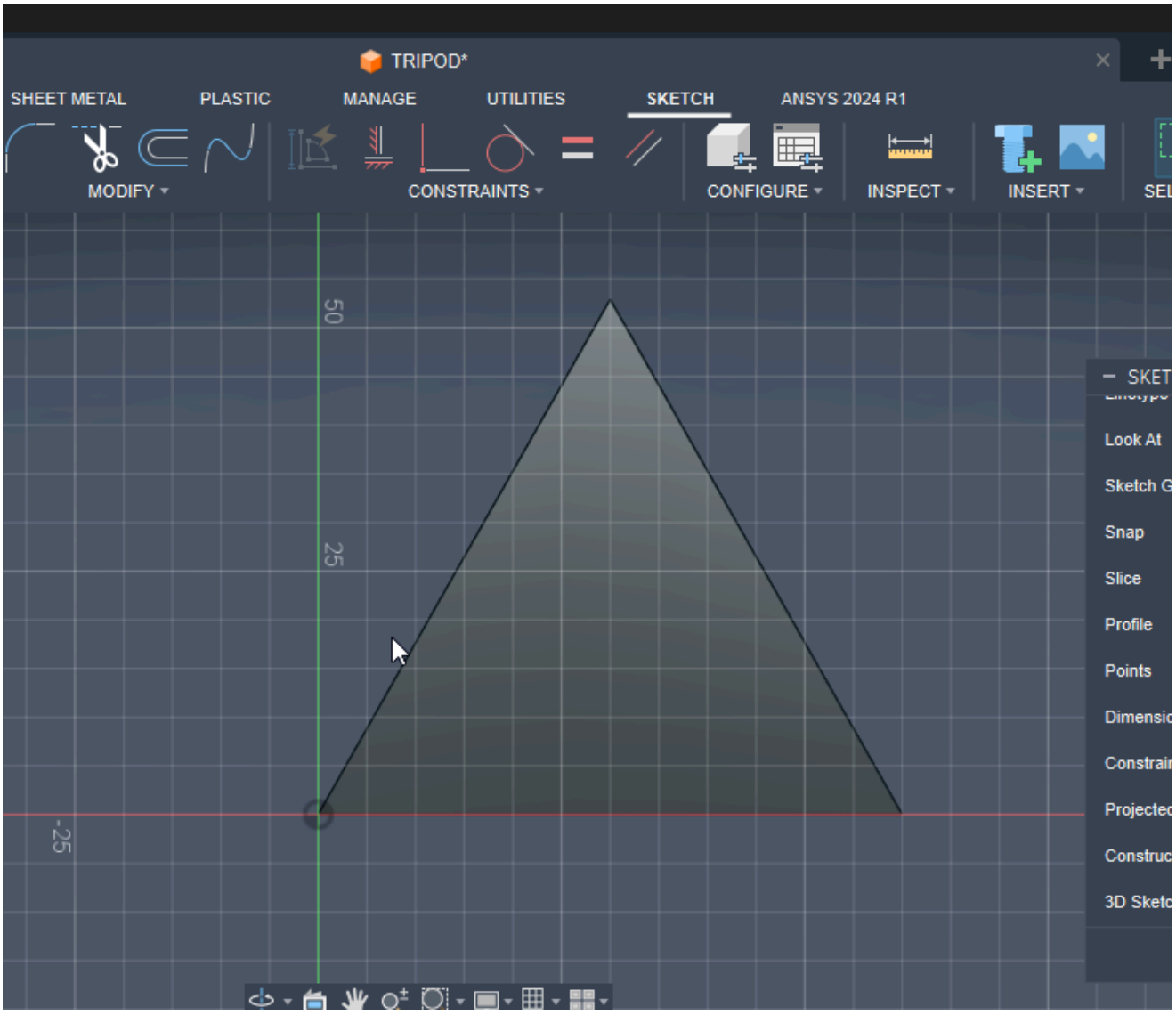
Place a point at the triangle centroid (intersection of construction lines).



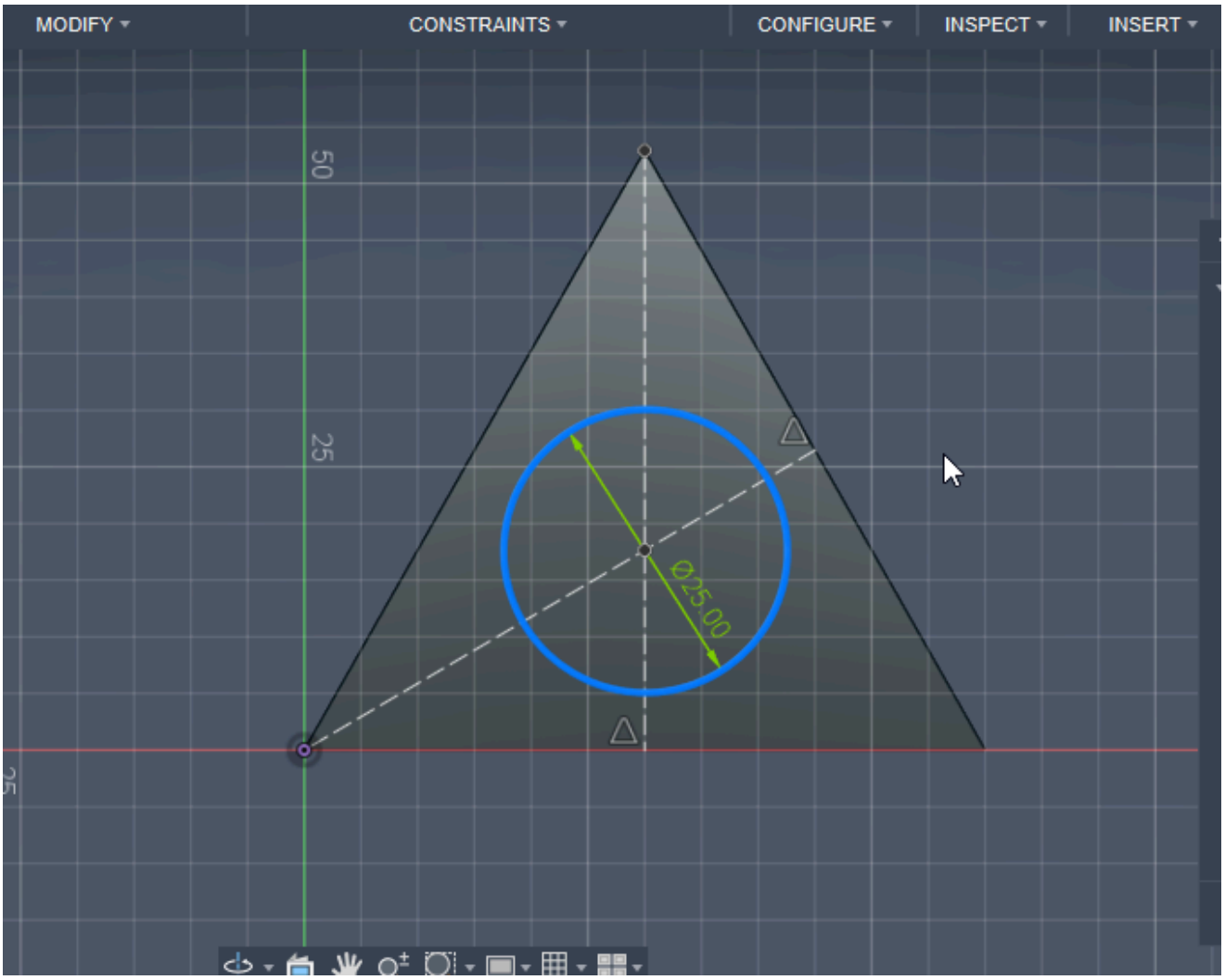
Use Loft — select the triangle profile and the center point to loft to a point (pyramid form).



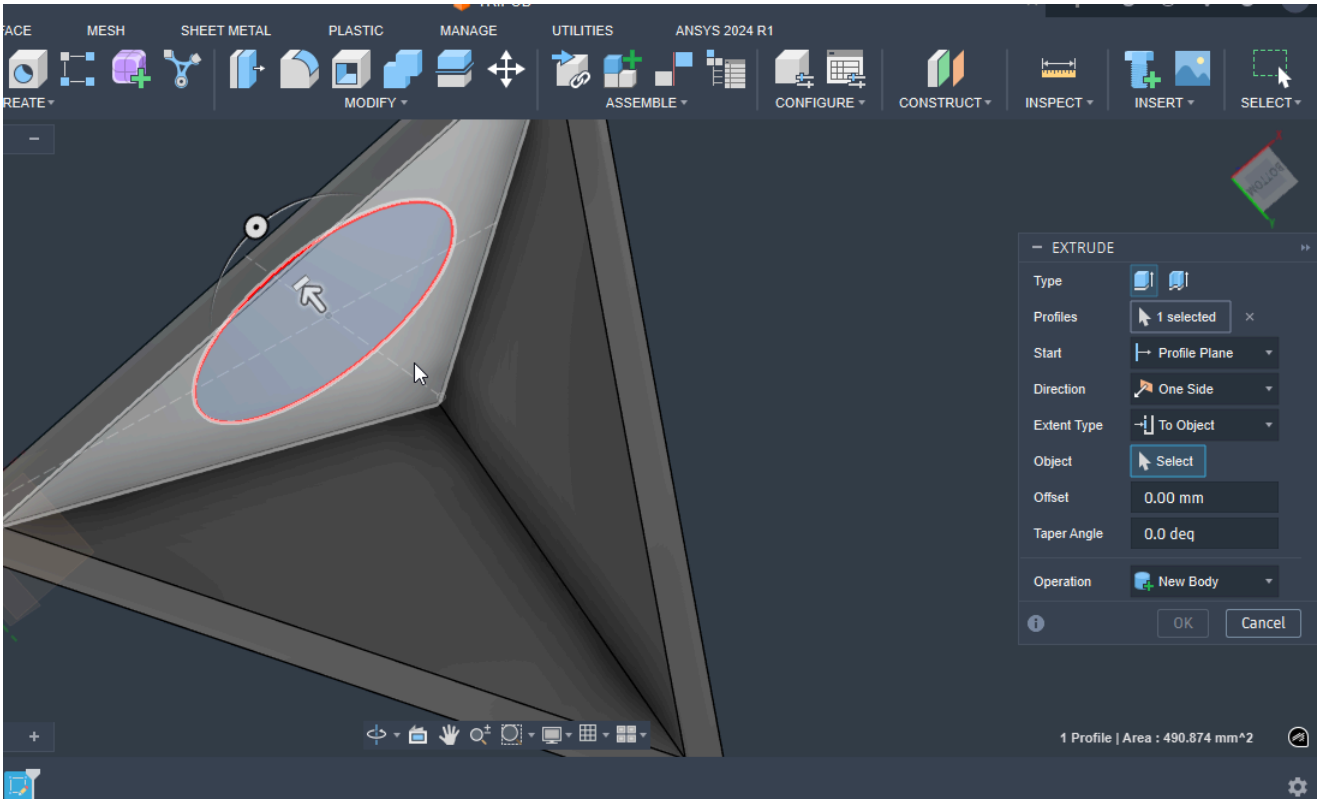
Apply Shell: set thickness = 2.5 mm to hollow the pyramid.



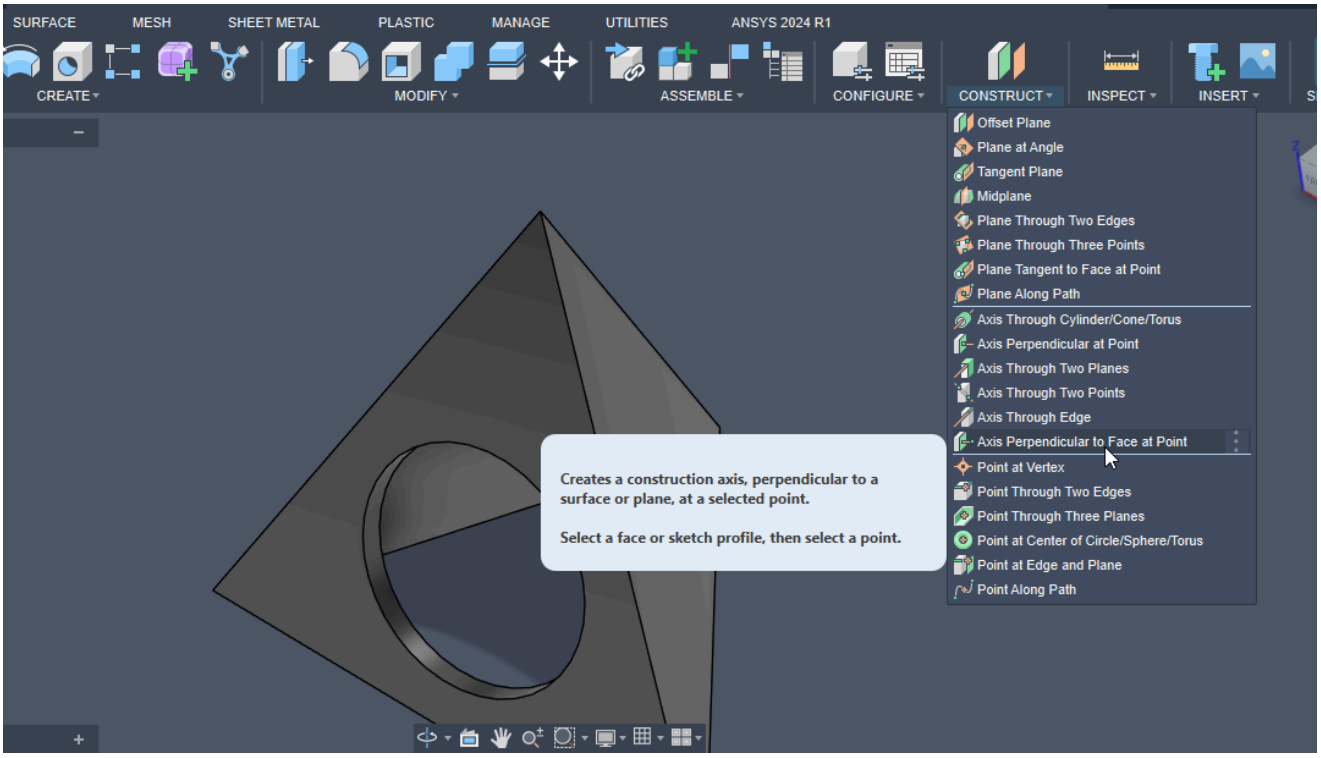
Create a new sketch on one of the pyramid faces (for hole pattern).



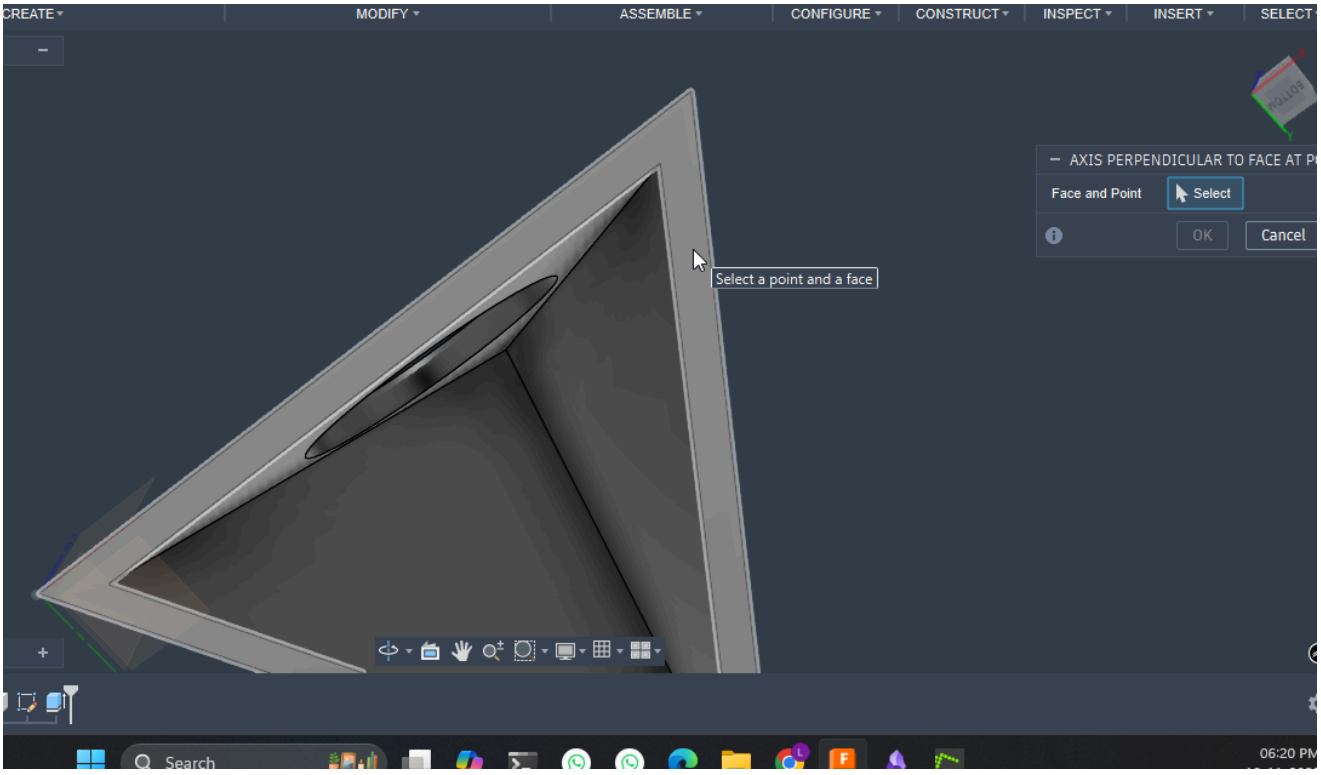
Draw a circle on that face (centered using projected geometry).



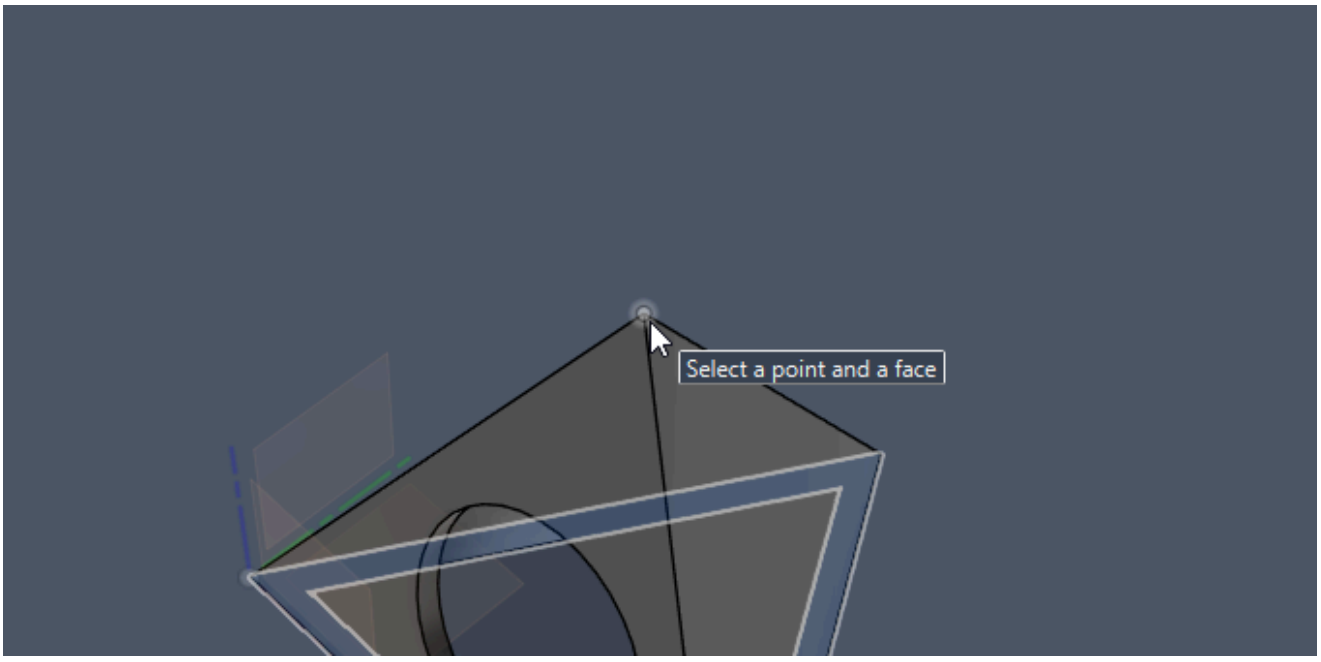
Use Extrude “To Object” (select inner face) to cut through to the shell interior cleanly.



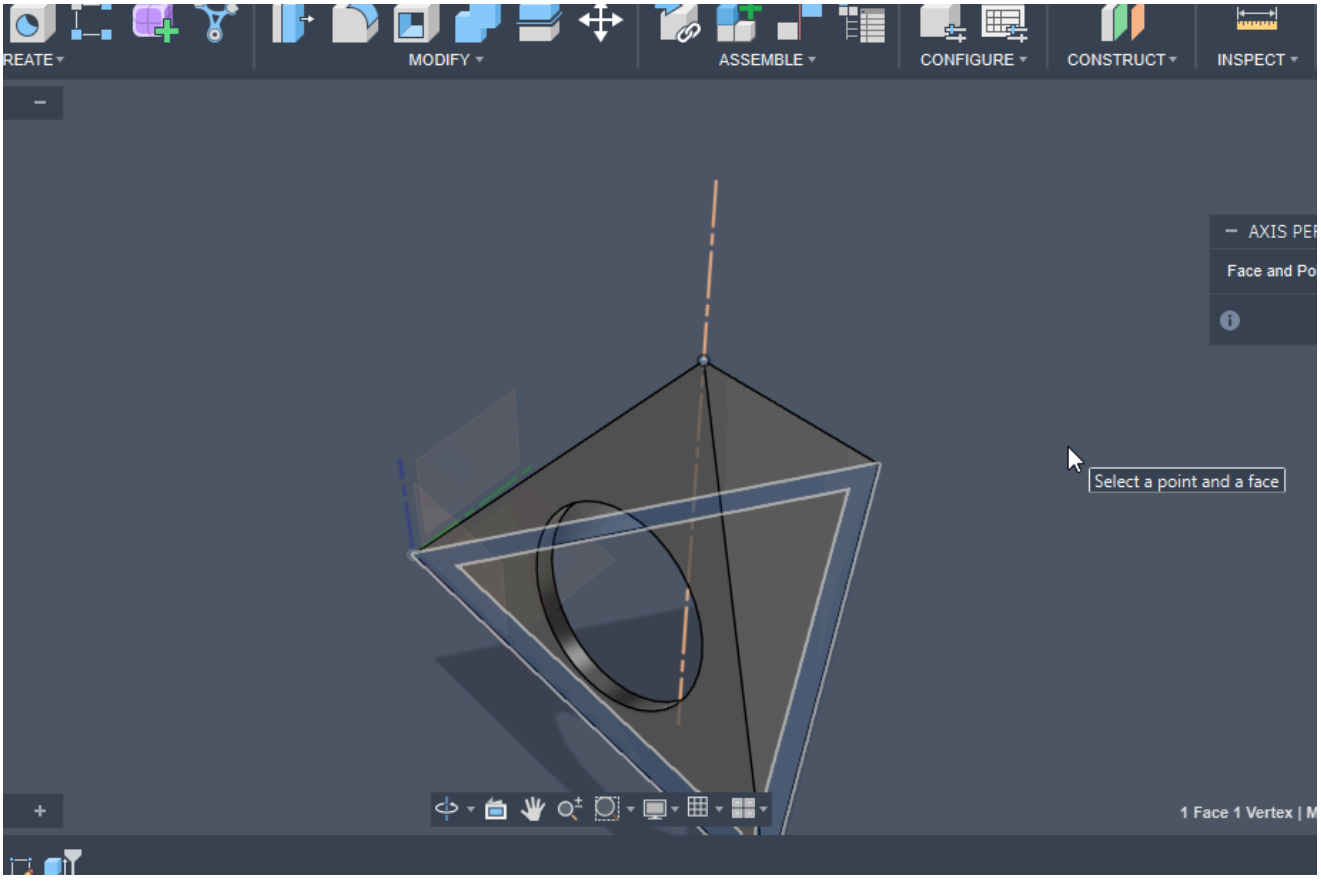
Create an axis perpendicular to the face at the circle center (Axis Perpendicular to Face at Point).



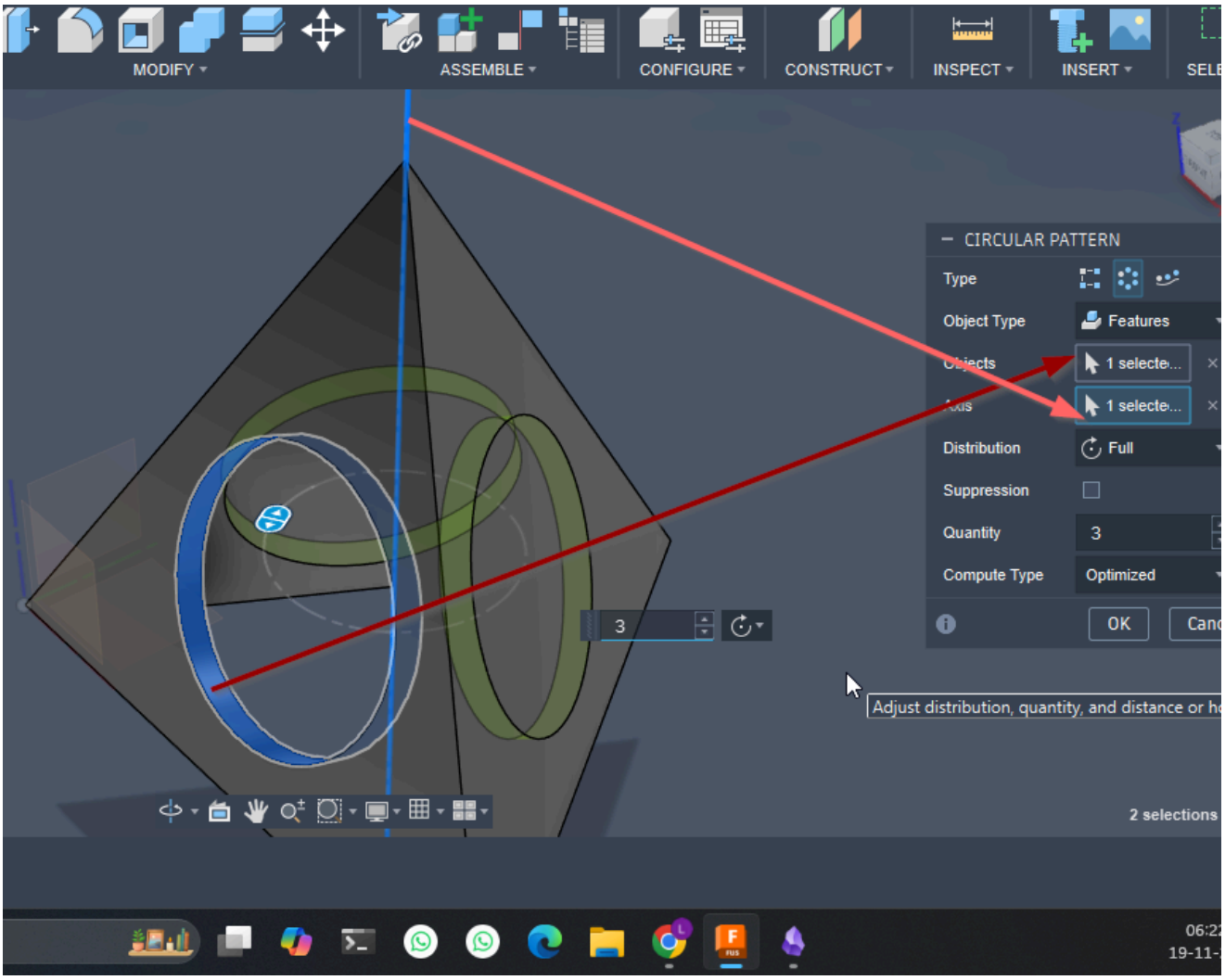
Select the face for the axis definition.



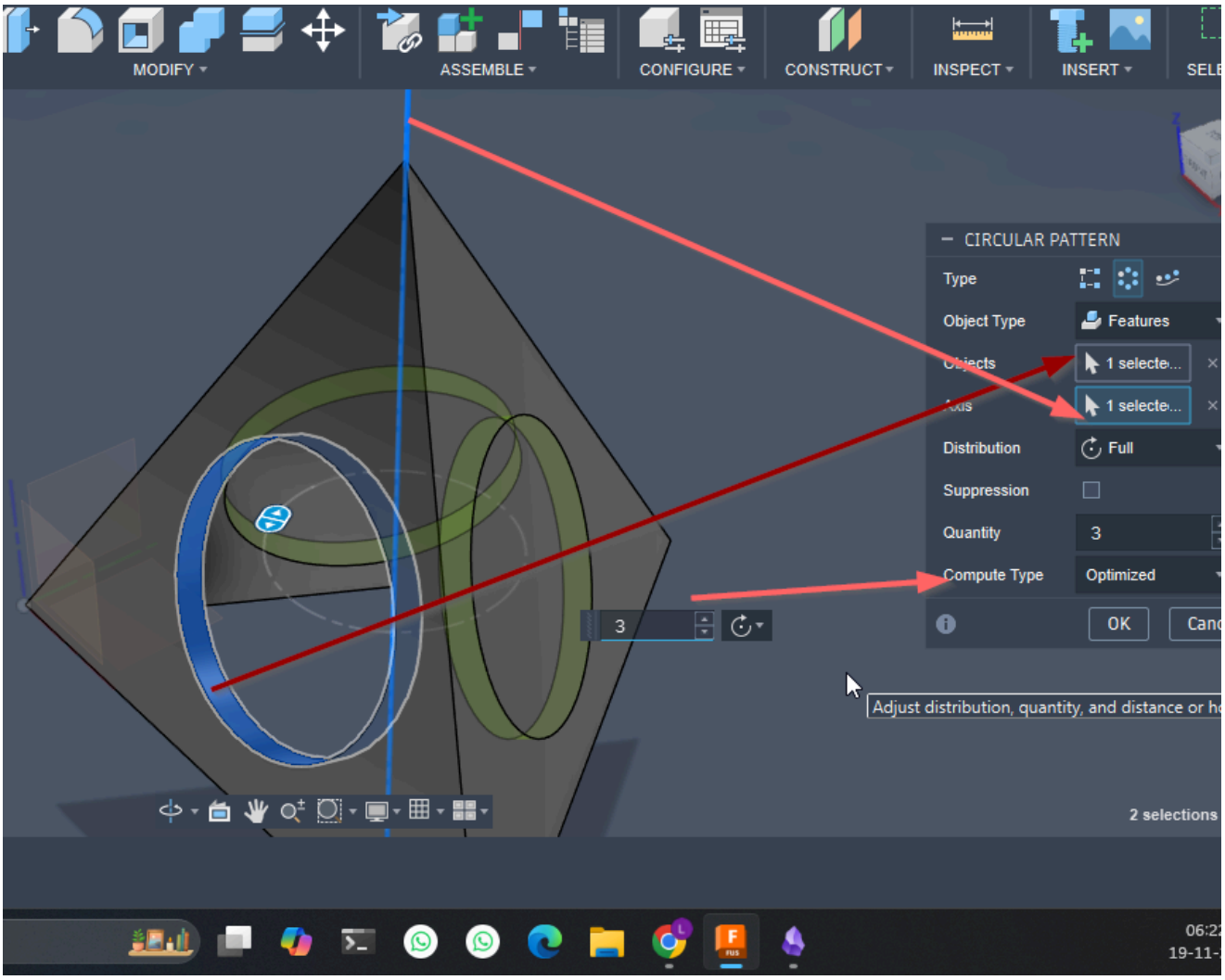
Select the point (circle center) to place the axis.



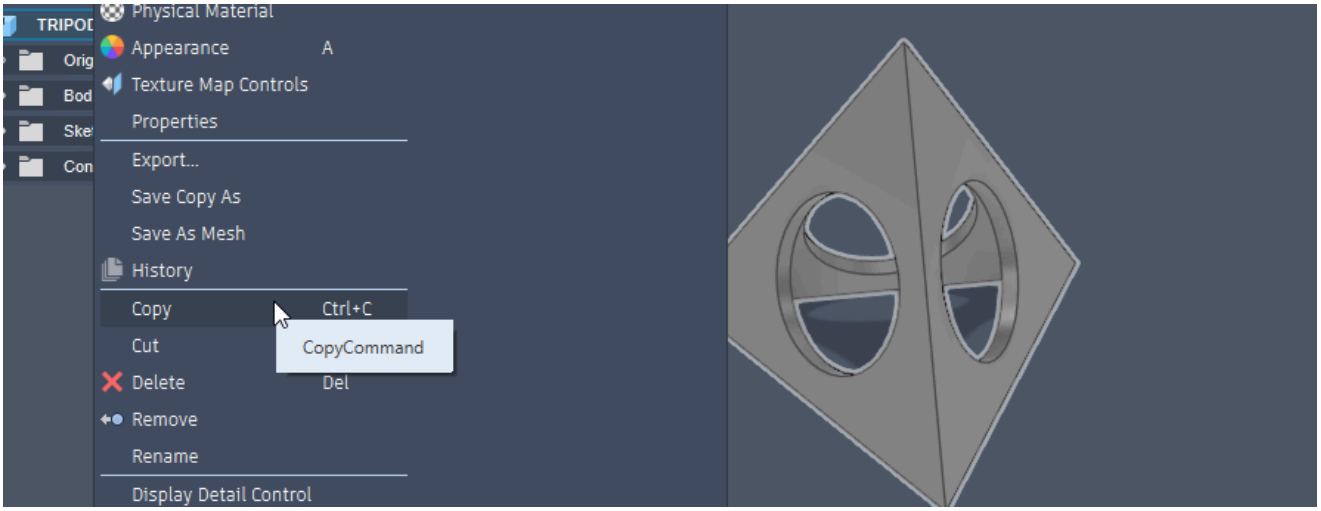
Axis drawn — ready for Circular Pattern.



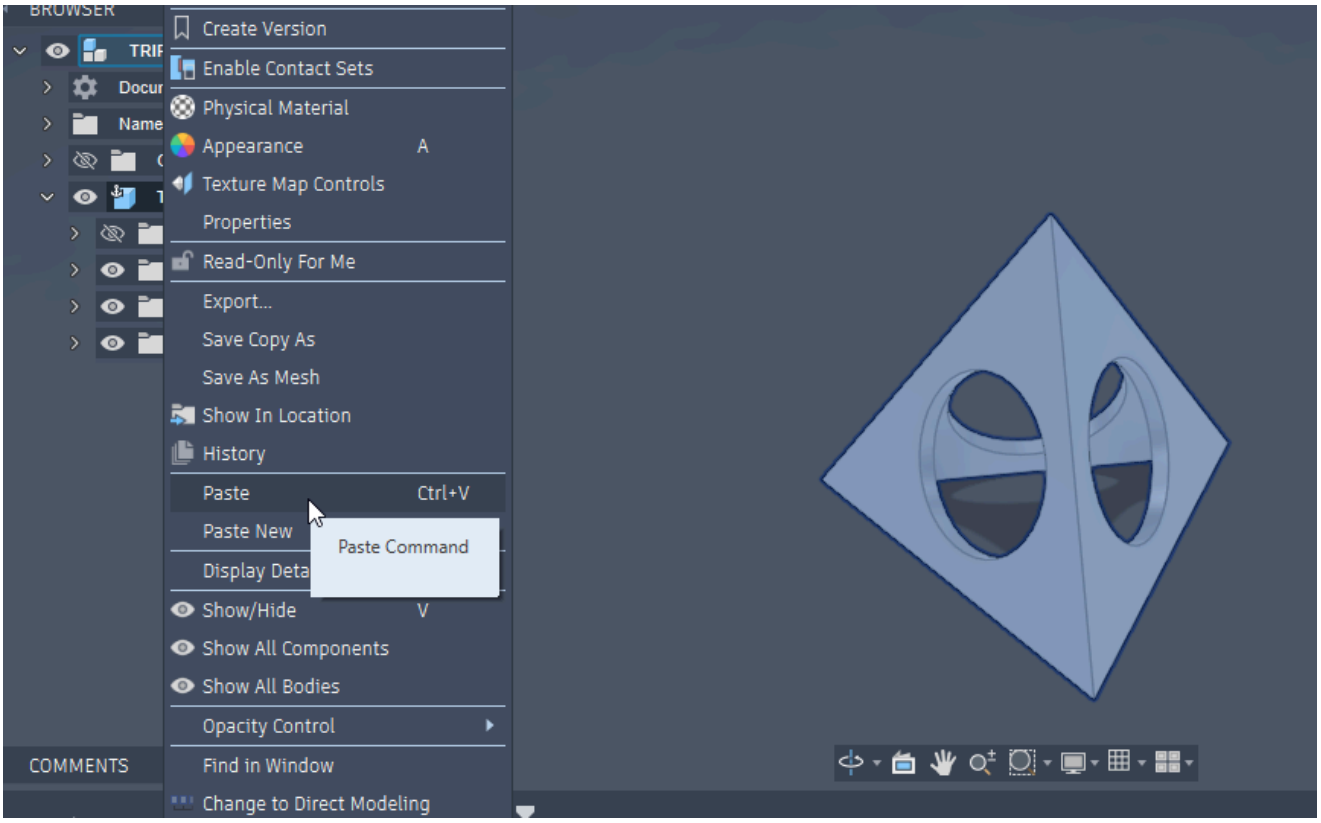
Open Circular Pattern (Type: Features / Object Type appropriate).



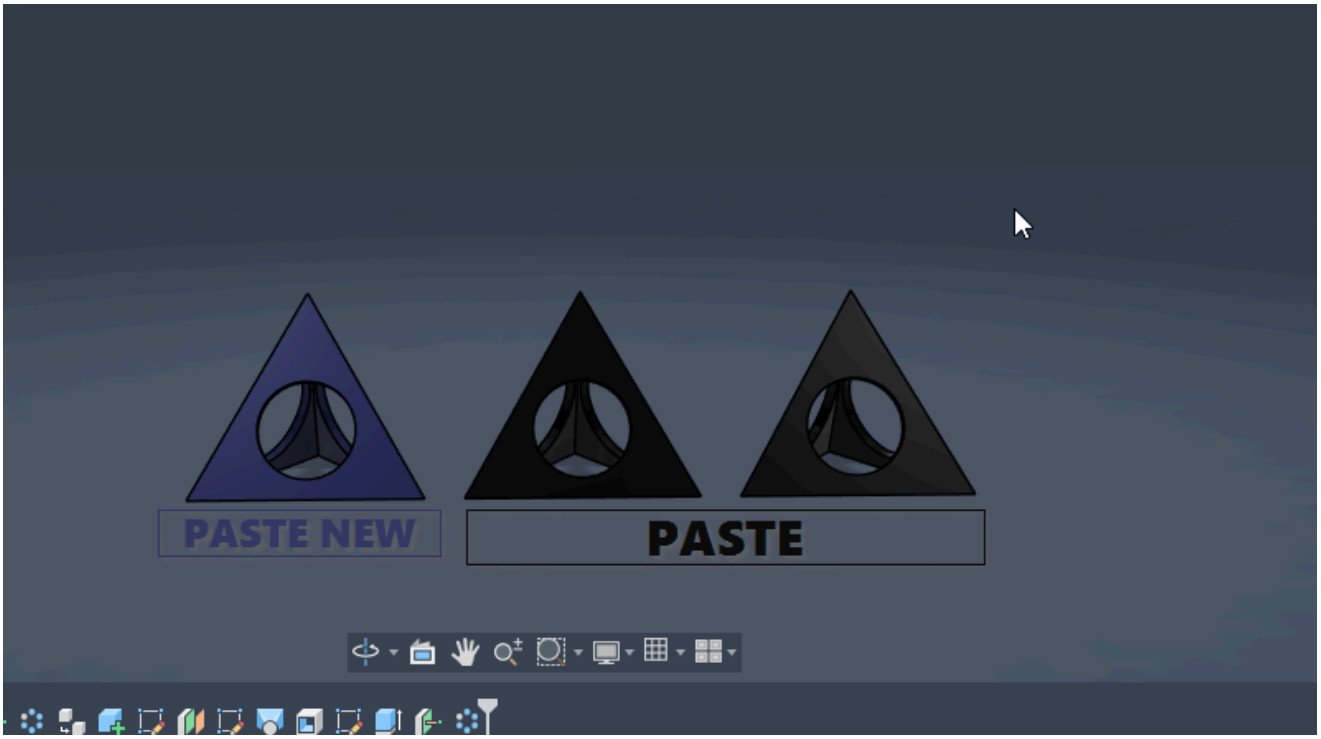
Set axis = the axis you created; set distribution = Full; set Quantity = 3 (or as needed).



Choose Compute Type carefully (Adjust / Optimized / etc.) — change if results look wrong.



Copy the Component (use Copy command in the Browser).



Two paste options: *Paste* vs *Paste New*.

- **Paste** — creates a linked copy (edits to one update the other).
- **Paste New** — creates an independent copy (edits do not affect the original).
